1 Vector Spaces

There are many sets for which it makes sense to add elements and multiply them by elements of F. *Vector spaces* are defined by abstracting the properties of the two operations on F^n we defined last time.

Definition 1.1 (Vector space)**.** A *vector space over F* is a set V together with an operation

$$\begin{aligned} V \times V &\to V \\ (u,v) &\mapsto u+v \end{aligned}$$

called *vector addition* (or just *addition*) and an operation

$$\begin{aligned} F \times V &\to V \\ (\alpha,v) &\mapsto \alpha v \end{aligned}$$

called *scalar multiplication*, which satisfy the same properties as those of F^n (which we call the *vector space axioms*):

Axioms of Addition
(A1) (Commutativity) $u+v=v+u$ for all $u,v \in V$.
(A2) (Associativity) $(u+v)+w=u+(v+w)$ for all $u,v,w \in V$.
(A3) (Additive Identity) There exists an element $0 \in V$ such that $0+v=v$ for every $v \in V$.
(A4) (Additive Inverses) To each $v \in V$ there exists an element $-v \in V$ such that $v+(-v)=0$.

Axioms of Scalar Multiplication
(S1) (Associativity) $(\alpha\beta)v=\alpha(\beta v)$ for all $\alpha,\beta \in F$, $v \in V$.
(S2) (Distributivity over scalar addition) $(\alpha+\beta)v=\alpha v+\beta v$ for all $\alpha,\beta \in F$, $v \in V$.
(S3) (Distributivity over vector addition) $\alpha(v+w)=\alpha v+\alpha w$ for all $\alpha \in F$, $v,w \in V$.
(S4) (Multiplicative identity) $1v=v$ for all $v \in V$.

In this context, elements of V are called *vectors* while elements of F are called *scalars*. The additive identity, 0, is called the *zero vector*. V is said to be a *complex vector space* when $F=\mathbb{C}$ and a *real vector space* when $F=\mathbb{R}$.

Example 1.2 (F^n)**.** We defined the notion of a vector space to have the same properties of addition and scalar multiplication as F^n, so F^n is clearly a vector space.

Example 1.3 (F^ω)**.** Define F^ω to be the set of all sequences of elements of F:

$$F^\omega := \{(x_1,x_2,\dots) : x_k \in F \text{ for } k=1,2,\dots\}.$$

Define addition and scalar multiplication on F^ω component-wise as for F^n:

$$\begin{aligned} (x_1,x_2,\dots)+(y_1,y_2,\dots) &:= (x_1+y_1,x_2+y_2,\dots) \\ \alpha(x_1,x_2,\dots) &:= (\alpha x_1,\alpha x_2,\dots). \end{aligned}$$

To show that these definitions make F^ω into a vector space, we need to check that the vector space axioms hold:
(A1):

$$\begin{aligned} x+y &= (x_1,x_2,\dots)+(y_1,y_2,\dots) \\ &= (x_1+y_1,x_2+y_2,\dots) \\ &= (y_1+x_1,y_2+x_2,\dots) \\ &= (y_1,y_2,\dots)+(x_1,x_2,\dots) \\ &= y+x \end{aligned}$$

(A2):

$$\begin{aligned} (x+y)+z &= (x_1+y_1,x_2+y_2,\dots)+(z_1,z_2,\dots) \\ &= ((x_1+y_1)+z_1,(x_2+y_2)+z_2,\dots) \\ &= (x_1+(y_1+z_1),x_2+(y_2+z_2),\dots) \\ &= (x_1,x_2,\dots)+(y_1+z_1,y_2+z_2,\dots) \\ &= x+(y+z) \end{aligned}$$

(A3):

$$(0,0,\dots)+(x_1,x_2,\dots) = (0+x_1,0+x_2,\dots) = (x_1,x_2,\dots)$$

shows that $0 = (0,0,\dots)$.

(A4):

$$(x_1,x_2,\dots)+(-x_1,-x_2,\dots) = (x_1-x_1,x_2-x_2,\dots) = (0,0,\dots),$$

which shows that for $x = (x_1,x_2,\dots)$, $-x = (-x_1,-x_2,\dots)$.

(S1):

$$\begin{aligned} (\alpha\beta)x &= (\alpha\beta)(x_1,x_2,\dots) \\ &= ((\alpha\beta)x_1,(\alpha\beta)x_2,\dots) \\ &= (\alpha(\beta x_1),\alpha(\beta x_2),\dots) \\ &= \alpha(\beta x_1,\beta x_2,\dots) \\ &= \alpha(\beta x) \end{aligned}$$

(S2):

$$\begin{aligned} (\alpha+\beta)x &= (\alpha+\beta)(x_1,x_2,\dots) \\ &= ((\alpha+\beta)x_1,(\alpha+\beta)x_2,\dots) \\ &= (\alpha x_1+\beta x_1,\alpha x_2+\beta x_2,\dots) \\ &= \alpha x+\beta x \end{aligned}$$

(S3):

$$\begin{aligned}\alpha(x+y) &= \alpha(x_1+y_1, x_2+y_2, \dots) \\ &= (\alpha(x_1+y_1), \alpha(x_2+y_2), \dots) \\ &= (\alpha x_1 + \alpha y_1, \alpha x_2 + \alpha y_2, \dots) \\ &= \alpha x + \alpha y\end{aligned}$$

(S4):

$$1x = 1(x_1, x_2, \dots) = (1x_1, 1x_2, \dots) = (x_1, x_2, \dots) = x.$$

Since all 8 axioms hold, F^{ω} is a vector space with respect to the operations defined above.

Example 1.4. $F^{m,n}$ Let $F^{m,n}$ denote the set of all $m \times n$ matrices with entries in F. For $A, B \in F^{m,n}$ and $\alpha \in F$, define

$$\begin{aligned}(A+B)_{ij} &:= A_{ij} + B_{ij} \\ (\alpha A)_{ij} &:= \alpha A_{ij}.\end{aligned}$$

The first equation says that the ij-entry of the matrix $A+B$ is defined to be equal to the sum of the ij-entries of A and B, while the second equation says the ij-entry of the matrix αA is defined to be the product of α and the ij-entry of A. For instance, if $m = 2, n = 3$, $\alpha = 2$,

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 4 & -2 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 3 & 5 & -4 \\ -8 & 2 & 1 \end{bmatrix},$$

then

$$A+B = \begin{bmatrix} 2+3 & -1+5 & 3-4 \\ 1-8 & 4+2 & -2+1 \end{bmatrix} = \begin{bmatrix} 5 & 4 & -1 \\ -7 & 6 & -1 \end{bmatrix}$$

and

$$2A = \begin{bmatrix} 2(2) & 2(-1) & 2(3) \\ 2(1) & 2(4) & 2(-2) \end{bmatrix} = \begin{bmatrix} 4 & -2 & 6 \\ 2 & 8 & -4 \end{bmatrix}.$$

These operations make $F^{m,n}$ into a vector space.

1. Verify that $F^{m,n}$ is a vector space with respect to the operations defined above.

2. Take $V = \{v\}$ (a singleton set). Define operations on V in such a way that makes V into a vector space. This is called the *zero vector space* (or *trivial vector space*).

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3. Can you define operations which make $\emptyset := \{\ \}$ (the empty set) into a vector space?

4. Let V be the set of positive real numbers, $\mathbb{R}^{>0}$, and define vector addition to be the ordinary multiplication of real numbers and scalar multiplication to be exponentiation. That is, for any $u, v \in \mathbb{R}^{>0}$ and $k \in \mathbb{R}$, define

$$u + v := uv$$
$$kv := v^k.$$

For example, with these definitions we have $2+3 := (2)(3) = 6$ and $(2)(3) = 3^2 = 9$. Show that this is a real vector space.

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Next we give some non-examples.

5. Let $V = F^2$. Define scalar multiplication as usual, but define vector addition as

$$(x_1, x_2) + (y_1, y_2) := (x_1 + x_2, y_1 + y_2 + x_1 y_1).$$

Explain why this is not a vector space. (Note that, since we are defining scalar multiplication as usual, we already know (S1)-(S4) hold, so we only need to check the axioms for vector addition.)

6. Let $V = F^2$. Define vector addition as usual, but define scalar multiplication by

$$\alpha(x_1, x_2) := (\alpha x_1, \alpha^2 x_2).$$

Explain why this is not a vector space.

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7. Let $V = \{(x,y) \in \mathbb{R}^2 : x \geq 0\}$ and define vector addition and scalar multiplication as the usual ones for $\mathbb{R}^2$.

$$(x_1,y_1) + (x_2,y_2) := (x_1 + x_2, y_1 + y_2)$$
$$k(x,y) := (kx,ky).$$

Explain why this is not a vector space.